

1 Reliable comparison of severe felt-intensity
2 classifiers with versioned and dependent event
3 data: a USGS DYFI benchmark

4 Akshay Sharma^{1*} and Lalji Prasad¹

5 ^{1*}Department of Computer Science and Engineering, SAGE University,
6 Indore, 452020, Madhya Pradesh, India.

7 *Corresponding author(s). E-mail(s): akssha74@gmail.com;
8 Contributing authors: lalji.research@gmail.com;

9 **Abstract**

10 Performance claims for engineering classifiers can change when the source cat-
11 alogue evolves, dependent events cross data splits, or small point differences
12 are interpreted without paired uncertainty. We ask whether performance and
13 model-superiority claims for severe felt-intensity classifiers remain supportable
14 when those risks are controlled. We release a versioned USGS “Did You Feel It?”
15 (DYFI) benchmark and a candidate-qualification workflow for applied machine-
16 learning engineers, benchmark maintainers, and reviewers. A candidate must
17 pass feature/proxy and role-isolation audits before temporal holdout scoring;
18 improvements are then assessed with proper scores and paired sequence-cluster
19 uncertainty. The frozen $M \geq 5$ snapshot contains 7,520 records, 2,362 eligible
20 events, and a 430-event temporal holdout. For a fixed random-forest split diag-
21 nostic, sequence grouping produced a Brier score 0.006 lower and an area under
22 the receiver-operating-characteristic curve 0.012 higher than random-event folds,
23 a small reversed contrast reported descriptively. All four admissible learned base-
24 lines improved on the no-skill reference, whereas the paired interval between the
25 two lowest-Brier models included zero. A worked qualification therefore supports
26 improvement over no-skill but rejects both a temporal-proxy candidate and a
27 false winner declaration. The resulting engineering test harness makes model
28 comparisons auditable across data version, split construction, uncertainty, and
29 software environment; it is not a real-time warning or operational triage system.

Keywords: model comparison, evaluation reliability, benchmark qualification,
dependent-event data, reproducibility, USGS DYFI

1 Introduction

Engineering model selection depends on comparisons that remain valid when a dataset is updated, observations are dependent, and leading scores are close. Without those controls, an apparent improvement can be caused by a changed population, a proxy for the outcome or evaluation axis, cross-split dependence, or unquantified sampling variation rather than by a better model. Standard benchmarking is valuable precisely because it fixes a common task and metrics; recent work in this journal, for example, compares object detectors on COCO to support model selection for engineering applications [29]. The reliability problem is harder for event catalogues because both the catalogue and the dependence structure can evolve.

The U.S. Geological Survey “Did You Feel It?” (DYFI) system is a useful engineering case study. It converts crowdsourced felt reports into a community-decimal intensity (CDI) for each earthquake and is widely reused as an interpreted data product [2, 22, 34, 35]. The target studied here is retrospective classification of whether event-maximum CDI reaches 6 from outcome-blind source metadata. It is not a real-time alert, causal model, or operational triage rule.

Our engineering question is: *do performance and model-superiority claims remain supportable when catalogue version, dependent-event splitting, temporal evaluation, paired uncertainty, and reconstruction are controlled?* The intended users are applied machine-learning engineers evaluating a candidate classifier, maintainers publishing a stable benchmark, and reviewers auditing a claimed improvement. They need two decisions: whether a candidate is admissible for comparison and whether its improvement over a declared reference is supported rather than a point-score fluctuation.

Existing felt-intensity studies use random or stratified partitions and do not provide that reusable decision contract [15, 20].

1.1 Contributions

We address unreliable model comparison by releasing a versioned benchmark and an executable candidate-qualification contract. Concretely, the work provides:

1. **A single-source, version-frozen DYFI benchmark** for the task “outcome-blind source metadata $\rightarrow$ later event-maximum $\max\text{CDI} \geq 6$ ”, frozen to one dated access instant so that any later catalogue revision defines a new version rather than a silent overwrite (Section 3).
2. **An executable six-category leakage audit** that rejects direct-label, post-outcome-ascertainment, identifier-proxy, and downstream-product features, and rejects cross-role sequence straddles and temporal-cutoff violations, accepting *only* the exact source-only feature set; it is exercised by 110 unit tests and rejects every deliberately contaminated test case (Section 4.1).
3. **Internally specified, outcome-independent roles** built on order-stable space-time sequence grouping and a temporal external holdout, so that split membership never uses the label (Section 4.2).
4. **A fixed, non-selective baseline suite** with calibration-aware proper scoring, Brier-score sequence-cluster bootstrap uncertainty, explicit handling of ties and unavailable calibration estimates, and single-use authorization of the decision-bearing protected evaluation (Sections 4.4–4.5).
5. **A candidate-qualification workflow** that fails closed on forbidden features or invalid roles and supports an improvement only when paired Brier-score uncertainty excludes zero (Section 5.5).

Table 1 Engineering failure modes addressed by the benchmark qualification contract

Failure source	Invalid engineering conclusion	Qualification control
Catalogue revision	Non-identical event population	Frozen access instant + SHA-256 release
Outcome-derived field	Label leakage	F1/F2 schema audit
Identifier or downstream proxy	Proxy leakage	F3/F4 field audit
Sequence crosses roles	Dependent-event contamination	F5 sequence-role audit
Temporal-cutoff violation	Future information in development	F6 temporal audit
Point-score ranking	False model-superiority claim	Paired sequence-cluster interval
Environment drift	Irreproducible scores/artifacts	Pinned environment + reconstruction hash

6. **A quantified split-sensitivity diagnostic** (naive random versus sequence-grouped splitting), together with a reproducible release bundle and machine-readable verification record (Section 5).

The learned models are worked candidates, not proposed production systems. Their comparison demonstrates the decisions the contract permits: all admissible learned candidates improve over no-skill, one temporal-proxy candidate is rejected, and the two lowest-Brier models remain unresolved.

1.2 Engineering significance and intended use

The output is an engineering test harness for model comparison. A future candidate enters through the released schema, is rejected if it exposes outcome/axis information or violates role isolation, is scored on the locked temporal holdout, and is compared with paired sequence-cluster uncertainty. This workflow prevents an engineer or reviewer from declaring superiority from a different catalogue version, a contaminated split, or an unresolved point difference. Table 1 names the failure prevented by each control, and Fig. 1 shows the resulting decision path.

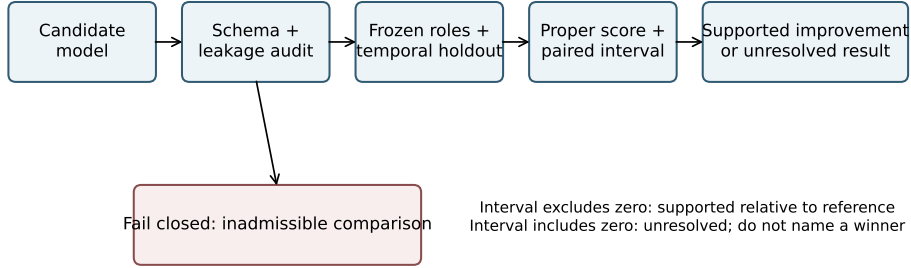


Fig. 1 Candidate-model qualification. A candidate that fails the schema, proxy, role, or temporal audit is inadmissible. An admitted candidate is scored on the fixed holdout and compared with paired uncertainty; an interval including zero remains unresolved rather than becoming a winner claim

2 Related work and novelty

DYFI and macroseismic intensity.

The DYFI system and its CDI construction, outlier handling, and post-event revision behaviour are documented in the foundational and review literature [2, 22, 34, 35]; these establish that CDI is an after-the-event, response-driven quantity subject to ascertainment effects, which is precisely why our protocol uses response count only to determine eligibility and forbids it as a predictive feature (Section 3). Rösler et al. [24] further show specifically for DYFI that interpreted products can change and should be preserved with timestamps for reproducible analysis. The dated, hashed snapshot released here implements that preservation principle. We reuse these provenance facts rather than re-deriving them.

Domain benchmark resources.

The INGe resource links Italian macroseismic intensities to colocated strong-motion records for benchmarking intensity–ground-motion relationships [18]. It is a domain-near resource precedent, but differs from the present event-level USGS DYFI

109 classification task, source-only feature boundary, leakage audit, and protected tem-
110 poral evaluation. The 2026 USGS Western North America resource contains about
111 696,000 site-level macroseismic observations from about 5,600 earthquakes and is pre-
112 dominantly DYFI [6]. It is a larger observational resource, but does not define this
113 event-maximum binary task, executable leakage audit, or frozen temporal evaluation.
114 Stockman et al. [28] provide the nearest broader leakage-aware seismological bench-
115 mark precedent: EarthquakeNPP standardizes public catalogues, repairs an improper
116 earthquake-forecasting split, compares neural point processes with ETAS, and finds
117 no consistent neural-model advantage. Its target is occurrence forecasting with chrono-
118 logical likelihood/CSEP evaluation, rather than source metadata predicting a later
119 citizen-reported felt-intensity endpoint; the resources are therefore methodologically
120 adjacent but task-distinct.

121 ***Engineering benchmark and model-comparison precedents.***

122 Comparative engineering studies use a fixed task and common metrics to turn model
123 outputs into selection evidence. Tian et al. [29], for example, compare object detec-
124 tors on COCO to support model choice for smart-city applications. In this journal,
125 machine-learning comparisons have also been tied to explicit engineering decisions in
126 rock-property estimation and pavement maintenance [1, 17], while seismic engineering
127 studies state the response quantity and design interpretation affected by the analy-
128 sis [26]. The present contribution follows that decision-oriented form but evaluates
129 the reliability of the comparison itself: whether a candidate is admissible, whether it
130 improves over a reference, and whether the evidence supports naming a winner.

131 An adjacent event-level binary study used post-event LastQuake felt reports to
132 classify high- versus low-impact earthquakes and evaluated later 2022 events [14]. Its
133 crowdsourced predictors, impact endpoint, and rapid-response objective differ from
134 the source-only DYFI benchmark and reusable evaluation contract developed here.

135 ***Macroseismic databases and citizen observations.***

136 Event-specific analysis in the 2019 Athens earthquake combined online felt reports
137 with ground motion and source modelling [12]. Crowdsourced macroseismic observa-
138 tions have also been used to characterize geological settings and spatial variations in
139 shaking response [25]. Curated macroseismic resources such as the Belgian traditional
140 macroseismic database and MACROISDATA preserve historical intensity observa-
141 tions for transparent reuse and later analysis [16, 27]. The present work extends this
142 seismological resource tradition from data preservation to a versioned, executable
143 contract for model evaluation.

144 ***Machine learning on felt/macroseismic intensity.***

145 Recent studies predict macroseismic intensity with random forests and gradient boost-
146 ing from instrumental or catalogue features [15, 20]. Machine learning has also been
147 applied to earthquake alert-level classification using intensity-related features and to
148 waveform detection with openly released seismic datasets [5, 19]. Those tasks use post-
149 event intensity descriptors or waveforms, rather than the source-only feature boundary
150 studied here. Each felt-intensity model application is evaluated under a random or
151 stratified partition and does not release a reusable, event-grouped, leakage-audited
152 benchmark; Mori and Naso [15] moreover predict from instrumental ground-motion
153 parameters rather than outcome-blind catalogue metadata, and Patelli et al. [20]
154 evaluate a single-system train/test split on seismometric inputs. Their concrete draw-
155 back for our purpose is the evaluation contract: a random or stratified split does not
156 remove within-sequence near-duplicate contamination, and none ships an executable
157 leakage audit or a version-frozen resource that a third party can rerun. Our protocol
158 addresses that specific drawback with internally specified sequence-grouped roles and
159 an executable six-category audit.

160 ***DYFI-based intensity models and concurrent leakage-aware work.***

161 Closest to our exact data source, Karthik Kumar et al. [10] predict DYFI commu-
162 nity intensity from the same catalogue metadata (magnitude, depth, location) with a
163 random-forest regressor on Indian events under a single naive train/test split (reported
164 $R^2 = 0.829$); our resource does not fit a competing regressor but instead supplies
165 a versioned, digitally fingerprinted, event- and sequence-grouped benchmark with
166 an executable leakage audit and a protected decision-bearing evaluation authorized
167 once, in place of the naive split that setup uses. Concurrently, Li et al. [13] release
168 a DYFI-grounded benchmark that simulates Modified Mercalli Intensity (MMI) for
169 two earthquakes (2014 Napa and 2019 Ridgecrest) with large language models and
170 includes an explicit location-identifier leakage test; our resource differs in task and
171 construction—an outcome-blind source-metadata→later-severe binary classification
172 over a decade-scale $M \geq 5$ corpus with internally specified sequence-grouped roles,
173 rather than multimodal per-zip-code MMI simulation for two events—so the two are
174 complementary rather than duplicative.

175 ***Cross-validation for dependent data.***

176 That grouped or blocked cross-validation is the correct evaluation strategy for tem-
177 porally, spatially, or hierarchically structured data is established practice [11, 23]; we
178 do not claim this as novel.

179 ***Novelty delta.***

180 Conceding the above, the delta this resource supplies that prior work does not is the
181 *released, versioned, hashed, single-source DYFI benchmark* itself: an executable six-
182 category audit covering forbidden fields, proxy variables, and split contamination; an
183 internally specified, sequence-grouped role assignment with a temporal external hold-
184 out, a documented severe endpoint with threshold and threshold sensitivity, a fixed
185 non-selective baseline suite with a no-skill comparator, a calibration/uncertainty/tie

186 contract, an exact-reproduction verification, and a quantified naive-versus-grouped
187 split diagnostic. A current-year search located no released resource occupying this
188 exact combination; novelty is stated as a concrete delta to located work, with no
189 first/only superlative asserted.

190 **3 Data, provenance, and outcome-boundary** 191 **disclosure**

192 *Source and single-instant pinning.*

193 The corpus is derived entirely from USGS DYFI and its parent ANSS Comprehensive
194 Catalog (ComCat), retrieved through the International Federation of Digital Seis-
195 mograph Networks (FDSN) event web service [30, 33]. The acquisition envelope is
196 outcome-independent: $M \geq 5$, product type `dyfi`, event type `earthquake`, starttime
197 2015-01-01 to endtime 2025-01-01. FDSN defines the end bound as inclusive; no frozen
198 record occurs at that boundary, so the realized snapshot spans 2015–2024. Results are
199 ordered by time. The preliminary count request and the final data request use this
200 same envelope. Because the catalogue evolves, the downloaded data were frozen at
201 one access instant (2026-09-01T06:33:30Z) and assigned a SHA-256 digital fingerprint.
202 A preliminary count changed from 7,575 to 7,576 during the acquisition session, doc-
203 umenting one live-catalogue revision. The final downloaded snapshot contained 7,520
204 DYFI records, and a count requested 50 seconds later in the same acquisition session
205 also returned 7,520. Thus, the one-record change belongs only to the earlier prelim-
206 inary checks; it is not a discrepancy in the frozen analysis snapshot. Likewise, the
207 56-record difference between the last preliminary count and the final 7,520-record snap-
208 shot is not an analysis exclusion: the preliminary count had no paired payload, so its
209 membership difference cannot be decomposed and is retained only as catalogue-version

210 provenance. The near-contemporaneous final query and count agree exactly. The dig-
211 ital fingerprint of the downloaded bytes, rather than a preliminary count, defines this
212 resource version. The frozen snapshot contains 7,520 DYFI records (Table 2, Fig. 2).

213 ***Protocol chronology.***

214 The file records an internal 2026-09-01 freeze but was first publicly timestamped on
215 2026-09-19. We therefore describe it as an internally frozen protocol, not an externally
216 registered preregistration.

217 ***Retrospective eligibility and post-report ascertainment.***

218 DYFI intensity is an after-the-event quantity whose reliability grows with the number
219 of felt reports [22]. To keep the label meaningful we apply a data-informed eligibility
220 filter of at least ten responses per event. The response count is used *only* for eligibil-
221 ity and is explicitly forbidden as a predictive feature, because it is a post-outcome
222 ascertainment proxy; this forbiddance is enforced by the leakage audit (family F2,
223 Section 4.1). Eligibility yields 2,362 events with severe prevalence 0.298 (703 severe
224 events). Of these, 2,360 events are assigned to the internally specified analysis roles
225 (Section 4.2) and contain 702 severe events; the remaining two are temporal-straddle
226 exclusions, one of which is severe. That single severe straddle event accounts for the
227 difference between the eligible cohort (703 severe) and the analysis-assigned set (702
228 severe), so eligibility is not the same as analysis-role assignment (Table 2).

229 ***Endpoint.***

230 We label an event severe when its event-maximum CDI is at least 6.0 (intensity VI or
231 greater). We use 6.0 as the primary threshold and report sensitivity at 5.5, 6.0, and
232 6.5. We also examine a near-threshold band of ± 0.5 (Section 5.2).

233 ***Outcome-boundary disclosure and held-out semantics.***

234 Only corpus-level outcome-blind counts and inherited aggregate feasibility statistics
235 were exposed before construction; no per-event label informed the endpoint, split, or
236 tuning choices, and role membership is assigned by an internally specified, outcome-
237 independent rule that never uses the label. We therefore describe the external holdout
238 as *held out from model development with aggregate feasibility previously known*, and we
239 deliberately do *not* call it “untouched” or “confirmatory.” Two event identifiers were
240 prelisted for quarantine because their aggregate counts had been seen during source
241 verification; neither identifier occurs in the frozen snapshot, so the rule is non-operative
242 in version 1 and excludes no record. The complete list of protocol decisions is reported
243 in Section 5.8. No respondent-level personally identifying information is redistributed;
244 only event-level source metadata, the severe label, the eligibility-only response count,
245 identifiers, and role assignments are released. Snapshot-level provenance is supplied
246 by the activated source-locator manifest and source payload SHA-256; version 1 does
247 not claim per-event product-version hashes. The release bundle follows datasheet-
248 for-datasets documentation [8] and findable, accessible, interoperable, and reusable
249 (FAIR) data principles [36].

250 4 Benchmark design

251 4.1 Feature sets and the six-category leakage audit

252 The primary *source-only* feature set contains magnitude and depth. It is the only fea-
253 ture set evaluated on the temporal holdout. An *expanded-metadata* sensitivity analysis
254 (baseline B3) additionally uses latitude, longitude, and event year, so it is not eval-
255 uated where those variables would disclose the holdout axis. The audit accepts only
256 an approved feature set with the expected fields in the expected order; any extra or
257 unknown field is rejected.

258 The executable audit enforces six leakage families. Field-level: F1 direct label or
 259 its constituents; F2 post-outcome ascertainment or revision (including the response
 260 count, eligibility-only); F3 identifier or source proxy; F4 downstream-product proxy.
 261 Split-level: F5 sequence cross-role straddle; F6 temporal-cutoff violation. An axis-
 262 proxy check additionally rejects origin year under a temporal holdout and, when a
 263 geographic holdout is configured, latitude/longitude/region. The latter audit branch
 264 is provided for reuse but was not executed as an external evaluation in this study. This
 265 enforces the B3 restriction. The audit is useful independently of model performance:
 266 it must reject every deliberately contaminated test case and accept only the approved
 267 clean feature set. It is exercised by 110 unit tests across four suites, all passing, and
 268 accepted the source-only feature set in the frozen snapshot.

269 4.2 Sequence grouping and outcome-independent roles

270 Events are grouped into seismic *sequences* by order-stable single-link space–time
 271 connected components (100 km, 30 days); the sequence, not the event, is the unit
 272 of role assignment, so events linked by this fixed rule cannot straddle roles. This
 273 rule is a deterministic grouping heuristic, not a validated catalogue of all physical
 274 mainshock–aftershock relationships; unlinked true aftershocks may therefore remain
 275 in different roles. Roles are assigned by an internally specified, outcome-independent
 276 rule (fixed assignment key, temporal cutoff year 2023, development fraction 0.2, five
 277 cross-validation folds); the rule provably never reads the label. The external holdout
 278 contains events on or after 2023. The frozen internal protocol also named a secondary
 279 leave-region-out geographic evaluation, but that branch was not executed and no
 280 geographic role manifest is claimed. The reported geographic result instead summa-
 281 rizes the temporal-holdout predictions by 10° cell (Table 9); Table 8 records this
 282 deviation. Digital fingerprints of the completed temporal role assignments are pub-
 283 lished so that third parties can verify them. The partition is exhaustive and disjoint:

284 $1,573 + 357 + 430 + 2 = 2,362$ events, comprising 1,573 train, 357 development, 430
 285 external temporal holdout, and 2 temporal-straddle exclusions (Table 2, Fig. 2). Con-
 286 struction (train plus development) is 1,930 events across 1,530 sequences; the holdout
 287 is 430 events across 354 sequences.

288 4.3 Baseline suite and exclusions

289 The fixed suite is B0 no-skill (constant prevalence), B1 magnitude-only logistic regres-
 290 sion, B2 source-only logistic regression, B3 expanded-metadata logistic regression
 291 (used only in compatible sensitivity analyses), B4 random forest (scikit-learn; 21), and
 292 B5 XGBoost [4]. Each learned baseline is tuned *only* by the protocol-defined deter-
 293 ministic grouped cross-validation on the construction corpus over a fixed grid, named
 294 before evaluation, and carried unchanged to the protected holdout with frozen best
 295 parameters. A machine-readable fit log enforces a maximum of 120 model fits; the exe-
 296 cuted run used 91. Two comparators are excluded for protocol-defined, reproducible
 297 reasons recorded in the protocol-decision table (Section 5.8): B3 is not evaluated on
 298 the temporal holdout because its feature set carries origin year, a temporal proxy; and
 299 the Atkinson–Wald (2007) DYFI intensity-prediction equation [2] is excluded because
 300 it cannot be applied to the source-only feature set and tectonically heterogeneous
 301 population without inventing a site-distance covariate, a per-event tectonic-regime
 302 assignment, and a probability link.

303 4.4 Metrics, calibration, uncertainty, and ranking

304 The primary score is the Brier score [3]; secondary scores are log-loss, AUROC,
 305 and calibration intercept/slope [31]. For numerical stability, log-loss clips predicted
 306 probabilities symmetrically to $[10^{-15}, 1 - 10^{-15}]$ before taking logarithms. Brier-
 307 score uncertainty is quantified by a sequence-cluster bootstrap ($n_{\text{boot}} = 2000$, 95%

intervals) that resamples whole sequences rather than events [7], with a minimum-valid-replicate rule. Calibration and feasibility states are *derived*, not hard-coded: a constant-prevalence predictor has no identifiable calibration slope, so the table reports that value as unavailable. Pairwise ranking uses a paired sequence-cluster bootstrap Brier difference with an explicit tie probability; the table reports whether each 95% interval includes zero, together with a tie-aware joint ordering (Section 5.6). Because the intervals use 2,000 Monte Carlo draws, near-zero endpoints are interpreted cautiously rather than as stable ranking guarantees. The protocol requested intervals for every metric; only Brier-based intervals were retained, so other metrics are descriptive point estimates (Table 8).

4.5 Protected evaluation and reproducibility

The official decision-bearing protected evaluation was authorized once by an automated, globally consumed token; the one-shot wrapper refuses a second authorized run. After the result was fixed, the token-free deterministic analytic core—including holdout scoring—was replayed twice for verification, and both replays reproduced the recorded result fingerprint exactly. Thus, single-use refers to result authorization and decision-bearing use, not to the total number of post-result computations. The frozen holdout predictions were also recovered bit-for-bit from the released rows and frozen best parameters (reported in Section 5.8), under a pinned scientific-Python environment (NumPy, SciPy, scikit-learn, XGBoost; 4, 9, 21, 32). All fits, exclusions, subsets, cutoffs, and pairwise comparisons are reported regardless of outcome; the only permitted omissions are the protocol-defined exclusions above. Computation is local-only under a pinned environment.

Reproducible artwork and generative AI assistance.

All figures were generated with Matplotlib 3.11.1 directly from the frozen analysis assets; no manual image manipulation or AI-generated artwork was used. ChatGPT

334 assisted language editing, formatting, code review, literature discovery, critical con-
335 sistency review, and revision suggestions. It did not generate data, execute model fits,
336 or create the reported numerical results. The authors independently verified every
337 cited source, computation, interpretation, and assisted change; all scientific judge-
338 ments and final decisions remained with the authors, who retain full responsibility for
339 the manuscript and released artifacts.

340 5 Results

341 All result-bearing values below are generated from the frozen analysis outputs and
342 checked automatically against their source records; no result cell was hand-edited.

343 5.1 Cohort and roles

344 Table 2 and Fig. 2 give the cohort accounting. Of 7,520 frozen DYFI records, 2,362
345 events pass the ten-response eligibility filter (severe prevalence 0.298; 703 severe
346 events), partitioned into 1,573 train, 357 development, and a 430-event temporal hold-
347 out, with two temporal-straddle exclusions. Thus 2,360 of the 2,362 eligible events
348 receive analysis roles; they contain 702 severe events, and the 703rd severe event is
349 one of the two straddle exclusions. Two identifiers prelisted for quarantine are absent
350 from the frozen snapshot, so that safeguard removes no version 1 record. Construc-
351 tion severe prevalence is 0.285 and holdout prevalence is 0.351; the higher holdout
352 prevalence reflects the temporal transport shift and is reported, not adjusted.

353 5.2 Difficulty and endpoint sensitivity

354 Table 3 and the left panel of Fig. 3 show the endpoint is not an artifact of a single
355 threshold: as the severe cut moves from 5.5 to 6.5, holdout prevalence falls smoothly
356 (0.458, 0.351, 0.260) and the source-only logistic-regression Brier score and AUROC
357 move monotonically (Brier 0.240, 0.181, 0.143; AUROC 0.734, 0.786, 0.815). On the

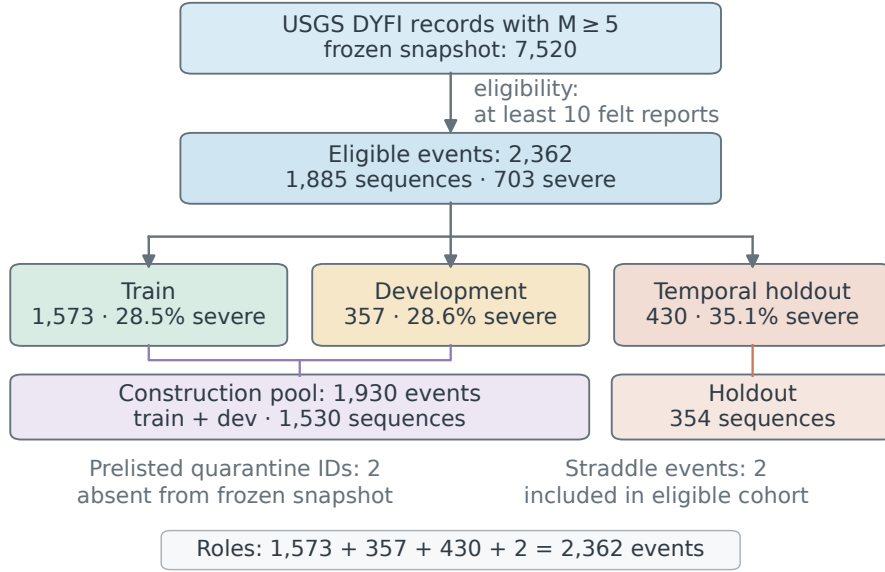


Fig. 2 Cohort construction and role assignment. On the frozen USGS DYFI $M \geq 5$ snapshot. The 7,520 source records yield 2,362 eligible events in 1,885 seismic sequences. Of these, 2,360 are assigned to train ($n = 1,573$), development ($n = 357$), or temporal holdout ($n = 430$); two straddle events are excluded from role assignment

358 construction corpus the near-threshold fraction ($|\max\text{CDI} - 6| < 0.5$) is 0.164 and
 359 the no-skill Brier score is 0.204, bounding the difficulty of the primary endpoint. This
 360 sensitivity check relabels and rescores the fixed B2 probabilities; it does not refit or
 361 recalibrate B2 at 5.5 or 6.5. The protocol-required aggregation-grid sensitivity was
 362 not executed and is disclosed as a deviation (Table 8).

363 5.3 Observed split-performance differences

364 The resource’s headline quantity is the optimism of naive random splitting relative
 365 to the sequence-grouped protocol, measured for a fixed leakage diagnostic—a random
 366 forest with 300 trees and unrestricted depth—under three fold-assignment schemes
 367 on the construction corpus (Table 4, Fig. 4). This diagnostic is distinct from the

Table 2 Cohort accounting and outcome-independent role assignment for the frozen USGS DYFI $M \geq 5$ snapshot.

Stage	Events	Severe	Severe rate	Sequences
USGS DYFI records with $M \geq 5$ (frozen snapshot)	7520	–	–	–
Eligible (at least 10 felt reports)	2362	703	0.298	1885
Train	1573	449	0.285	–
Development	357	102	0.286	–
Construction (train+dev)	1930	551	0.285	1530
External temporal holdout	430	151	0.351	354
Temporal-straddle excluded	2	1	–	1

Note. Roles were fixed without using outcomes. Two event identifiers prelisted for quarantine are absent from the frozen snapshot, so the safeguard removes no version 1 record. Severe = 1 when an event’s maximum community decimal intensity (CDI) is at least 6. The eligible severe count reconciles as 703 = 449 (train) + 102 (development) + 151 (holdout) + 1 (straddle exclusion). Thus the 2360 role-assigned events contain 702 severe events.

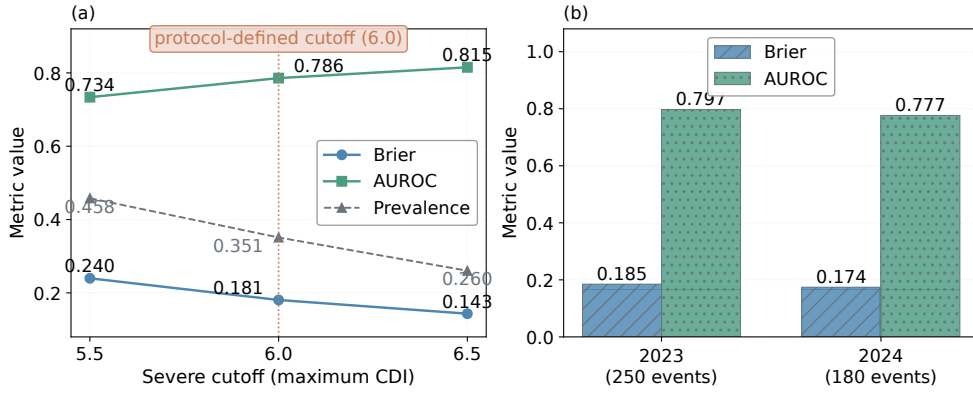


Fig. 3 Endpoint and temporal sensitivity. Results for the source-only logistic regression (B2). (a) Performance as the severe-intensity cutoff varies. (b) Performance within the 2023 ($n = 250$) and 2024 ($n = 180$) portions of the temporal holdout. The cutoff analysis rescores fixed B2 probabilities without refitting. Lower Brier score and higher AUROC are better

368 tuned temporal-holdout baseline B4, whose maximum depth is 3. On this $M \geq 5$ one-
369 row-per-event universe the point estimates are small and slightly reversed: sequence
370 grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than
371 random-event splitting, calculated from unrounded scores. The sequence-grouped split
372 therefore scores marginally *better* than the naive random split, not worse. The frozen
373 internal protocol required uncertainty reporting for this contrast but did not define
374 a paired interval estimator; the executed result contains point estimates only. We

Table 3 Sensitivity to the severe-intensity cutoff on the temporal holdout, evaluated with the source-only logistic regression (B2).

Maximum CDI cutoff	Severe rate	Brier score	AUROC
5.5	0.458	0.240	0.734
6.0 ^a	0.351	0.181	0.786
6.5	0.260	0.143	0.815

Note. ^a Protocol-defined primary cutoff (≥ 6). CDI denotes community decimal intensity. The sensitivity rows relabel and rescore fixed B2 probabilities; B2 is not refit or recalibrated for the alternate cutoffs. In the construction sample, 0.164 of events lie within 0.5 CDI units of the primary cutoff, and the no-skill Brier score is 0.204. Lower Brier score and higher AUROC indicate better performance.

Table 4 Cross-validated performance of a fixed leakage diagnostic (random forest: 300 trees, unrestricted depth) under three fold-assignment schemes in the construction sample.

Fold assignment	Brier score	AUROC	Log-loss	Folds
Random-event (naive)	0.2048	0.6952	0.9640	5
Event-grouped (control)	0.2009	0.7058	0.9223	5
Sequence-grouped (primary)	0.1990	0.7076	0.9709	5

Note. The diagnostic is distinct from the tuned temporal-holdout baseline B4 (depth 3). Using unrounded values, sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting. The frozen internal protocol required uncertainty but did not define the paired interval estimator; the executed report retained point estimates only. This deviation is disclosed, and no post-hoc interval or equivalence claim is added. Because there is one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation is different and can produce different scores.

disclose this unfulfilled element as a deviation and do not invent a post-hoc inter-
val. Accordingly, we report the contrast descriptively, not as evidence of equivalence,
a true zero effect, leakage, or a win. Its plausible structural context is that 1,885
sequences are largely singleton (1,530 in construction), leaving limited within-sequence
near-duplicate structure for a random split to leak. This does not prove a small true
effect. With one row per event, event grouping imposes the same grouping constraint
as random-event assignment, but its deterministic fold allocation differs and there-
fore produces different scores. It is reported as an intermediate control. The protocol
remains reusable on datasets with more multi-event sequences. The present result nei-
ther demonstrates practically important leakage from random splitting nor establishes
its absence.

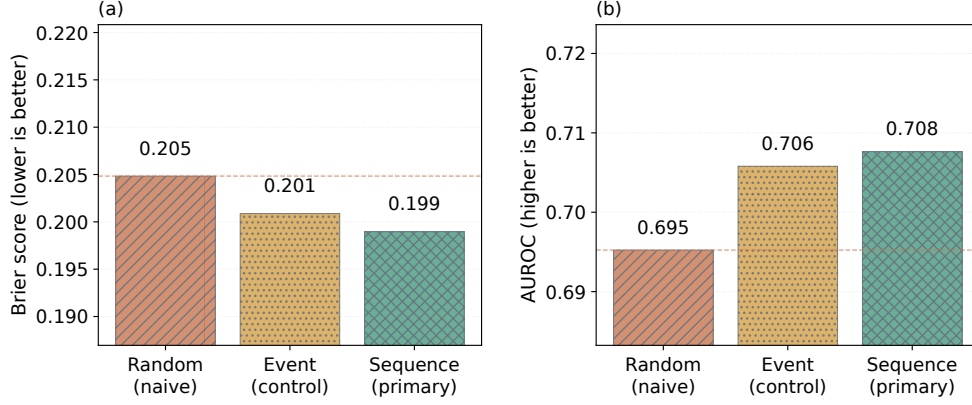


Fig. 4 Split-assignment diagnostic. Five-fold performance of the fixed diagnostic (random forest: 300 trees, unrestricted depth) under random-event, event-grouped, and sequence-grouped fold assignment in the construction sample. Relative to random-event splitting, sequence grouping produced (a) a Brier score 0.0059 lower and (b) an AUROC 0.0124 higher. The diagnostic is distinct from tuned temporal-holdout baseline B4. The small, reversed differences provide no evidence of performance inflation from random splitting in this sample; dashed lines mark the corresponding random-event score in each panel

5.4 Baseline performance and calibration

Table 5 and Fig. 5 give per-baseline holdout performance ($n = 430$) with sequence-cluster bootstrap 95% intervals. The no-skill B0 has Brier 0.232 ([0.212, 0.253]); its calibration slope cannot be estimated because it assigns the same probability to every event. The learned baselines have lower Brier (B1 0.198, B2 0.181, B4 0.187, B5 0.181) with overlapping intervals; calibration intercept/slope pairs are finite and positive. No baseline is claimed to win; the table reports proper scores, uncertainty, and derived calibration states only.

5.5 Worked candidate qualification

Table 6 applies the released engineering decision rule to the six fixed candidates. B3 fails before scoring because event year exposes the temporal evaluation axis. B1, B2, B4, and B5 pass the audit and each has a paired Brier interval below the no-skill B0 reference. The contract therefore supports their improvement over B0. It does not support naming B2 or B5 as the winner: their direct paired interval includes

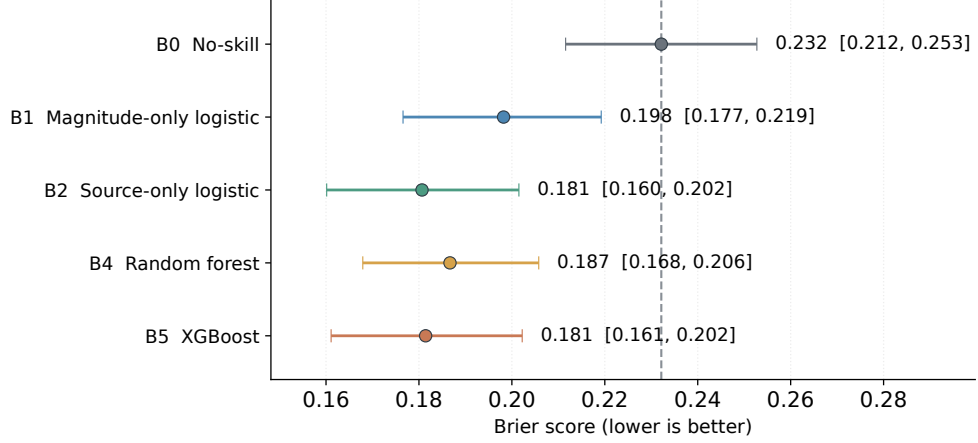


Fig. 5 Temporal-holdout baseline performance. Brier scores and 95% sequence-cluster bootstrap confidence intervals for the five baselines on the temporal holdout ($n = 430$). Lower values are better; the vertical dashed line marks the no-skill B0 score. B3 is omitted because it contains event year, which would reveal the temporal holdout period (Table 8)

Table 5 Per-baseline performance on the temporal holdout ($n = 430$) with sequence-cluster bootstrap 95% CIs and derived calibration states.

Baseline	Brier	95% CI	Log-loss	AUROC	Calibration ^a	Accuracy
B0 (no-skill)	0.232	[0.212, 0.253]	0.658	0.500	INFEASIBLE	0.65
B1 (magnitude-only logistic)	0.198	[0.177, 0.219]	0.583	0.719	0.32 / 1.02	0.70
B2 (source-only logistic)	0.181	[0.160, 0.202]	0.536	0.786	0.64 / 1.29	0.73
B4 (random forest)	0.187	[0.168, 0.206]	0.553	0.777	0.74 / 1.48	0.73
B5 (XGBoost)	0.181	[0.161, 0.202]	0.541	0.783	0.42 / 1.07	0.74

Note. ^a Intercept / slope. These values cannot be estimated for B0 because it assigns the same probability to every event. B3 includes event year and is therefore omitted from the temporal holdout (Table 8).

Accuracy uses a fixed probability cutoff of 0.5 that was not selected on the holdout. Log-loss clips probabilities to $[10^{-15}, 1 - 10^{-15}]$. Lower Brier score and log-loss are better; higher AUROC and accuracy are better. The two lowest-Brier models cannot be distinguished in their direct paired comparison.

400 zero. This worked example demonstrates the benchmark’s intended use—admit only
 401 valid candidates, support improvements with uncertainty, and leave an unresolved
 402 comparison unresolved.

403 5.6 Pairwise ranking stability

404 Table 7 reports all ten pairwise Brier-score comparisons. Every learned baseline per-
 405 forms better than the no-skill reference B0 (all four intervals exclude zero). Among
 406 the learned models, the B2–B4 and B2–B5 comparisons have intervals that include

Table 6 Worked candidate-model qualification using the frozen temporal holdout

Candidate	Audit gate	Holdout [95% CI]	Brier	Brier difference vs B0 [95% CI]	Contract deci- sion
B0 no-skill	Pass (reference)	0.232 0.253]	[0.212,	—	Reference only
B1 magnitude-only logistic	Pass	0.198 0.219]	[0.177,	-0.0339 [-0.0511, -0.0171]	Supported vs B0
B2 source-only logistic	Pass	0.181 0.202]	[0.160,	-0.0515 [-0.0681, -0.0342]	Supported vs B0
B4 random forest	Pass	0.187 0.206]	[0.168,	-0.0455 [-0.0598, -0.0307]	Supported vs B0
B5 XGBoost	Pass	0.181 0.202]	[0.161,	-0.0507 [-0.0682, -0.0319]	Supported vs B0
B3 expanded-metadata logistic	Fail	—	—	—	Inadmissible: event year reveals tempo- ral axis

Decision rule. A candidate is first admitted by the schema, proxy, role, and temporal audits. A lower paired Brier interval that excludes zero supports improvement over the declared reference; otherwise the comparison remains unresolved. All four learned baselines improve over B0. B2 versus B5 remains unresolved (B2–B5 -0.0008, 95% CI [-0.0078, 0.0068]), so the contract prevents a false winner declaration.

zero. The remaining learned-model intervals exclude zero, although the B4–B5 lower endpoint (0.0003) is near the Monte Carlo boundary. In the joint bootstrap ordering, B2 has the highest probability of being uniquely best (0.613), followed by B5 (0.387), but the direct B2–B5 comparison remains uncertain. These results therefore do not justify naming a winning model.

The frozen internal protocol allowed an “equivalent” verdict but did not specify a positive numerical equivalence margin; its resolution floor was zero. We therefore do not invent a post-hoc margin or label any models equivalent. Instead, we report each point difference, 95% interval, and whether that interval includes zero. Table 8 records this interpretive restriction, which changes no result.

5.7 Temporal and geographic subgroup summaries

Table 9 and the right panel of Fig. 3 report the temporal and geographic subgroup summaries for the source-only logistic regression within the temporal holdout. The two holdout years are internally consistent (2023: $n = 250$, Brier 0.185, AUROC 0.797; 2024: $n = 180$, Brier 0.174, AUROC 0.777). The geographic subgroup summary spans

Table 7 Pairwise comparison of baseline Brier scores on the temporal holdout using a paired sequence-cluster bootstrap.

Model A	Model B	A – B	95% CI	$P(\text{A better})$	CI relation
B0	B1	0.0339	[0.0171, 0.0511]	0.000	excludes 0
B0	B2	0.0515	[0.0342, 0.0681]	0.000	excludes 0
B0	B4	0.0455	[0.0307, 0.0598]	0.000	excludes 0
B0	B5	0.0507	[0.0319, 0.0682]	0.000	excludes 0
B1	B2	0.0176	[0.0101, 0.0249]	0.000	excludes 0
B1	B4	0.0115	[0.0023, 0.0205]	0.009	excludes 0
B1	B5	0.0168	[0.0055, 0.0273]	0.001	excludes 0
B2	B4	-0.0060	[-0.0118, 0.0002]	0.970	includes 0
B2	B5	-0.0008	[-0.0078, 0.0068]	0.613	includes 0
B4	B5	0.0052	[0.0003, 0.0099]	0.017	excludes 0

Note. Lower Brier score is better, so a negative A – B difference favors model A. Intervals use 2,000 Monte Carlo bootstrap draws; near-zero endpoints are interpreted cautiously and not as stable ranking guarantees. Because no numerical equivalence margin was specified before analysis, intervals including zero are not labelled equivalent. B0 is the no-skill reference; B1 is magnitude-only logistic regression; B2 is source-only logistic regression; B4 is random forest; and B5 is XGBoost.

82 occupied 10° cells (per-cell Brier 0.002–0.825, median 0.151). These values stratify the same 430 temporal-holdout predictions; they are not a leave-region-out evaluation and support no geographic or cross-system generalization claim.

5.8 Exclusions, deviations, and reproducibility

Table 8 explains the protocol decisions most relevant to interpretation: excluded comparators and records, the unexecuted geographic transport branch, the unfulfilled split-contrast interval, and the absence of an equivalence claim. The tagged public release contains the complete SHA-256 fingerprints, software environment, and machine-readable verification details. Its public-release manifest fingerprints every distributed file. Three original execution-bookkeeping records are self-referential—they cannot contain their own final fingerprints—and are therefore reported separately rather than treated as analysis-file mismatches.

Table 8 Excluded analyses and disclosed deviations affecting interpretation.

Decision	Treatment	Rationale
B3 expanded-metadata model	Omitted from temporal holdout	Includes event year, which reveals the temporal evaluation period.
Published DYFI intensity equation	Omitted as comparator	Needs unavailable site distance, tectonic-region inputs, and probability conversion.
Two prelisted quarantine IDs	No effect in version 1	Both are absent from the frozen snapshot; no released record is excluded.
Geographic transport evaluation	Not executed	Only temporal-holdout geographic subgroups are reported; no leave-region-out or generalization claim.
Aggregation-grid sensitivity	Not executed	Required but absent; only threshold sensitivity of fixed predictions is reported.
Uncertainty coverage	Partial deviation	The protocol requested intervals for every metric and the split contrast; only Brier-based intervals were retained.
Equivalence interpretation	Descriptive only	No numerical margin was specified; intervals are reported without equivalence labels.

Note. These decisions were recorded before, or disclosed with, the analysis. No result, subset, cutoff, or comparison was omitted because of its observed outcome.

Table 9 Temporal and geographic subgroup summaries within the temporal holdout for the source-only logistic regression (B2).

Subset	Events	Brier score	AUROC
Temporal 2023	250	0.185	0.797
Temporal 2024	180	0.174	0.777
Geographic cells within temporal holdout (82)	430	0.151 (med.)	–

Note. The geographic row partitions the same 430 temporal-holdout predictions into 82 occupied 10° cells. It is a descriptive subgroup summary, not a leave-region-out or external geographic evaluation. The entry is the median cell-level Brier score; cell values range from 0.002 to 0.825 and are included in the public release.

6 Discussion

Engineering use and decision consequence.

The resource is designed for an engineer or reviewer assessing a new event-level classifier. The user does not choose a convenient split or reconstruct a moving dataset; the candidate is evaluated against a released contract. The audit first rejects forbidden

439 or axis-revealing inputs, the role manifest prevents linked events from crossing the
440 declared boundary, and the paired Brier interval determines whether improvement
441 over a reference is supported. The worked example changes two decisions: B3 is inad-
442 missible before holdout scoring, while B2 and B5 are both supported over no-skill but
443 cannot be ranked decisively. Without the contract, the same point scores invite an
444 unsupported winner claim.

445 *Interpreting the split contrast.*

446 The observed point estimate is small and slightly reversed. In these folds the naive
447 random split did not inflate the diagnostic score, but the missing interval pre-
448 vents an equivalence, zero-effect, or “little practical effect” inference. The protocol
449 remains useful for datasets containing more multi-event sequences; this result neither
450 demonstrates leakage nor supports a win.

451 *No winning model.*

452 Every learned baseline separates from the no-skill reference. However, the direct B2–
453 B5 comparison between the two lowest-Brier models is unresolved, as is B2–B4, so
454 the results do not establish a decisive winner. Consistent with the resource’s scope,
455 we claim no winning model and no operational, real-time, lead-time, cross-system, or
456 causal benefit.

457 **7 Limitations and threats to validity**

458 *Outcome and catalogue limitations.*

459 The endpoint is magnitude-dominated, CDI remains sensitive to responder coverage,
460 and magnitude and depth are frozen preferred catalogue values rather than guaranteed
461 first-alert values. The ten-response rule mitigates ascertainment; response count is
462 used only for eligibility. B1 provides the simple magnitude comparator.

463 ***Unexecuted sensitivity and transport branches.***

464 The geographic values are temporal-holdout subgroups, not external transport evi-
465 dence. The protocol-defined geographic evaluation and aggregation-grid sensitivity
466 were not executed, and no alternative to the fixed 100-km/30-day sequence defini-
467 tion was tested. The rule guarantees co-assignment only for events it links; it does
468 not guarantee identification of every physical aftershock relationship. Consequently,
469 no geographic, aggregation, sequence-definition, or complete aftershock-isolation
470 robustness claim is made.

471 ***Inference and held-out semantics.***

472 The protocol-required split-contrast uncertainty interval was not retained, so that
473 result remains descriptive with no equivalence or zero-effect inference. Non-Brier
474 scores remain descriptive point estimates. Pairwise intervals use 2,000 Monte Carlo
475 draws and near-zero endpoints are interpreted cautiously. The temporal holdout was
476 excluded from model development, but aggregate feasibility was previously known; it
477 is neither “untouched” nor confirmatory.

478 ***Public release.***

479 Version 1 of the data, code, frozen results, and reproduction instructions
480 is available in a tagged public repository at [https://github.com/akssha74/](https://github.com/akssha74/dyfi-benchmark-reliability)
481 [dyfi-benchmark-reliability](https://github.com/akssha74/dyfi-benchmark-reliability). The release retains complete digital fingerprints so that
482 later revisions remain distinguishable from the version evaluated here.

483 **8 Conclusion**

484 Reliable engineering model selection requires more than reporting the lowest point
485 score. The released DYFI benchmark makes a candidate pass an auditable schema
486 and role contract, evaluates it on a fixed temporal holdout, and attaches paired uncer-
487 tainty to the comparison decision. In the worked qualification, four learned candidates

improve over no-skill, B3 is rejected as a temporal proxy, and the two lowest-Brier models remain unresolved; the contract therefore prevents an inadmissible comparison and a false winner declaration. The random-versus-sequence diagnostic is small and reversed in this sparse-sequence corpus, so it supplies no evidence of random-split optimism. All results reconstruct from the frozen release. The contribution is a reusable test harness for trustworthy model comparison, not a real-time warning, causal model, or operational decision system.

Declarations

- **Funding.** This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.
- **Competing interests.** The authors have no relevant financial or non-financial interests to disclose.
- **Ethics approval.** Not applicable. The study analyzes public, aggregate event-level USGS products rather than individual questionnaire responses; it involved no interaction with participants, identifiable data, human-subject collection, animals, or biological material.
- **Consent to participate.** Not applicable; no participants were recruited.
- **Consent for publication.** Not applicable.
- **Availability of data and materials.** Version 1 data, results, manifests, and instructions are available under CC0 1.0 at <https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.1.0>. Source data are USGS DYFI and ANSS ComCat (public domain; [10.5066/F7J101C8](https://doi.org/10.5066/F7J101C8) and [10.5066/F7MS3QZH](https://doi.org/10.5066/F7MS3QZH)) and are reconstructable via the released fetch-and-verify source-locator manifest. No respondent-level personally identifying information is redistributed.
- **Availability of code.** Project: DYFI Benchmark Reliability. Homepage: <https://github.com/akssha74/dyfi-benchmark-reliability>. Archived version: release v1.1.0.

514 Operating systems: platform independent. Programming language: Python. Licence:
515 MIT. Restrictions on non-academic use: none. Full prediction reconstruction and
516 model tests use the separately pinned scientific-Python environment.

517 • **Author contributions.** A.S. conceived and designed the benchmark protocol,
518 developed the software, curated the data, performed the analyses, validated the
519 results, and drafted the manuscript. L.P. contributed to study design and inter-
520 pretation, supervised the work, and critically revised the manuscript for important
521 intellectual content. Both authors reviewed and approved the final manuscript and
522 agree to be accountable for all aspects of the work.

523 • **Acknowledgements.** We thank the USGS DYFI and ANSS Comprehensive
524 Catalog programs.

525 References

- 526 [1] Alnaqbi A, Zeiada W, Al-Khateeb G (2025) A hybrid machine learning method of
527 support vector regression with particle swarm optimization for predicting IRI in
528 continuously reinforced concrete pavement. Journal of Engineering and Applied
529 Science 72:128. <https://doi.org/10.1186/s44147-025-00706-9>
- 530 [2] Atkinson GM, Wald DJ (2007) “Did you feel it?” intensity data: A surpris-
531 ingly good measure of earthquake ground motion. Seismological Research Letters
532 78(3):362–368. <https://doi.org/10.1785/gssrl.78.3.362>
- 533 [3] Brier GW (1950) Verification of forecasts expressed in terms of probability.
534 Monthly Weather Review 78(1):1–3. [https://doi.org/10.1175/1520-0493\(1950\)](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2)
535 078<0001:VOFEIT>2.0.CO;2

- 536 [4] Chen T, Guestrin C (2016) XGBoost: A scalable tree boosting system. In:
537 Proceedings of the 22nd ACM SIGKDD International Conference on Knowl-
538 edge Discovery and Data Mining, pp 785–794, [https://doi.org/10.1145/2939672.](https://doi.org/10.1145/2939672.2939785)
539 [2939785](https://doi.org/10.1145/2939672.2939785)
- 540 [5] Ertuncay D, de Lorenzo A, Costa G (2025) Deep learning based earthquake and
541 vehicle detection algorithm. *Journal of Seismology* 29(1):269–281. [https://doi.](https://doi.org/10.1007/s10950-024-10267-8)
542 [org/10.1007/s10950-024-10267-8](https://doi.org/10.1007/s10950-024-10267-8), first published online in 2024
- 543 [6] Fathian Sabet A, Abrahamson N, Parker GA, et al (2026) Macroseismic Intensity
544 Dataset in Western North America. U.S. Geological Survey data release, [https:](https://doi.org/10.5066/P143CZ3R)
545 [//doi.org/10.5066/P143CZ3R](https://doi.org/10.5066/P143CZ3R), URL <https://doi.org/10.5066/P143CZ3R>
- 546 [7] Field CA, Welsh AH (2007) Bootstrapping clustered data. *Journal of the Royal*
547 *Statistical Society: Series B (Statistical Methodology)* 69(3):369–390. [https://doi.](https://doi.org/10.1111/j.1467-9868.2007.00593.x)
548 [org/10.1111/j.1467-9868.2007.00593.x](https://doi.org/10.1111/j.1467-9868.2007.00593.x)
- 549 [8] Gebru T, Morgenstern J, Vecchione B, et al (2021) Datasheets for datasets.
550 *Communications of the ACM* 64(12):86–92. <https://doi.org/10.1145/3458723>
- 551 [9] Harris CR, Millman KJ, van der Walt SJ, et al (2020) Array programming with
552 NumPy. *Nature* 585:357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- 553 [10] Karthik Kumar P, Vismay, Apoorva, et al (2025) Machine learning based DYFI
554 intensity prediction for seismic events in India. In: 2025 9th International Confer-
555 ence on Electronics, Communication and Aerospace Technology (ICECA). IEEE,
556 <https://doi.org/10.1109/iceca66444.2025.11382872>
- 557 [11] Kaufman S, Rosset S, Perlich C, et al (2012) Leakage in data mining: Formulation,
558 detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*
559 6(4):15:1–15:21. <https://doi.org/10.1145/2382577.2382579>

- [12] Kouskouna V, Ganas A, Kleanthi M, et al (2021) Evaluation of macroseismic intensity, strong ground motion pattern and fault model of the 19 july 2019 mw5.1 earthquake west of athens. *Journal of Seismology* 25(3):747–769. <https://doi.org/10.1007/s10950-021-09990-3>
- [13] Li L, Li D, Ou Z, et al (2025) LLMs as world models: Data-driven and human-centered pre-event simulation for disaster impact assessment. In: *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Suzhou, China, pp 3078–3096, <https://doi.org/10.18653/v1/2025.emnlp-main.153>, released DYFI-grounded MMI benchmark with a location-identifier leakage test
- [14] Lilienkamp H, Bossu R, Cotton F, et al (2023) Utilization of crowdsourced felt reports to distinguish high-impact from low-impact earthquakes globally within minutes of an event. *The Seismic Record* 3(1):29–36. <https://doi.org/10.1785/0320220039>
- [15] Mori F, Naso G (2025) An explainable machine learning model for predicting macroseismic intensity for emergency management. *Remote Sensing* 17(10):1754. <https://doi.org/10.3390/rs17101754>
- [16] Neefs B, Van Noten K, Vanneste K, et al (2025) The belgian traditional macroseismic (BTM) database of the twentieth century. *Journal of Seismology* 29(5):1005–1036. <https://doi.org/10.1007/s10950-024-10266-9>, first published online in 2024
- [17] Niu L, Cui Q, Luo J, et al (2024) Unconfined compressive strength prediction of rock materials based on machine learning. *Journal of Engineering and Applied Science* 71:137. <https://doi.org/10.1186/s44147-024-00472-0>

- 584 [18] Oliveti I, Faenza L, Michelini A (2022) INGe: Intensity-ground motion data set
585 for Italy. *Annals of Geophysics* 65(1):DM102. <https://doi.org/10.4401/ag-8709>
- 586 [19] Özüpak Y (2026) Explainable and statistically validated machine learning for
587 earthquake alert-level classification using seismic intensity features. *Journal of*
588 *Seismology* 30(5). <https://doi.org/10.1007/s10950-026-10444-x>
- 589 [20] Patelli L, Cameletti M, De Rubeis V, et al (2026) Machine learning for prompt
590 estimation of macroseismic intensity from seismometric data in Italy. *Scientific*
591 *Reports* 16:7265. <https://doi.org/10.1038/s41598-026-35740-x>
- 592 [21] Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine
593 learning in Python. *Journal of Machine Learning Research* 12:2825–2830
- 594 [22] Quitoriano V, Wald DJ (2020) USGS “Did you feel it?”—science and lessons
595 from 20 years of citizen science-based macroseismology. *Frontiers in Earth Science*
596 8:120. <https://doi.org/10.3389/feart.2020.00120>
- 597 [23] Roberts DR, Bahn V, Ciuti S, et al (2017) Cross-validation strategies for
598 data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*
599 40(8):913–929. <https://doi.org/10.1111/ecog.02881>
- 600 [24] Rösler B, Stein S, Hough SE (2022) On the documentation, independence, and
601 stability of widely used seismological data products. *Frontiers in Earth Science*
602 10:988098. <https://doi.org/10.3389/feart.2022.988098>
- 603 [25] Saraò A, Tamaro A, Sandron D, et al (2023) On the crowdsourcing of macro-
604 seismic data to characterize geological settings. *International Journal of Disaster*
605 *Risk Reduction* 96:103934. <https://doi.org/10.1016/j.ijdr.2023.103934>

- 606 [26] Seleemah MA, Helam MS, Abu-alenein MA, et al (2022) Response of Aswan cable-
607 stayed bridge to spatial non-synchronous seismic excitations. Journal of Engineer-
608 ing and Applied Science 69:70. <https://doi.org/10.1186/s44147-022-00124-1>
- 609 [27] Sira C, Schirr L, Bertrand V, et al (2025) MACROSIDATA: Safeguarding
610 and promoting the heritage of french macroseismic surveys carried out between
611 1921 and 1996. Journal of Seismology 29(5):1145–1160. <https://doi.org/10.1007/s10950-025-10322-y>
612
- 613 [28] Stockman S, Lawson DJ, Werner MJ (2026) EarthquakeNPP: A benchmark for
614 earthquake forecasting with neural point processes. Transactions on Machine
615 Learning Research URL <https://openreview.net/forum?id=dIcNAg6ZuZ>, article
616 6105
- 617 [29] Tian J, Jin Q, Wang Y, et al (2024) Performance analysis of deep learning-
618 based object detection algorithms on COCO benchmark: A comparative study.
619 Journal of Engineering and Applied Science 71:76. <https://doi.org/10.1186/s44147-024-00411-z>
620
- 621 [30] U.S. Geological Survey, Earthquake Hazards Program (2017) Advanced national
622 seismic system (ANSS) comprehensive catalog of earthquake events and products.
623 U.S. Geological Survey, <https://doi.org/10.5066/F7MS3QZH>
- 624 [31] Van Calster B, McLernon DJ, van Smeden M, et al (2019) Calibration: The
625 Achilles heel of predictive analytics. BMC Medicine 17:230. <https://doi.org/10.1186/s12916-019-1466-7>
626
- 627 [32] Virtanen P, Gommers R, Oliphant TE, et al (2020) SciPy 1.0: Fundamental
628 algorithms for scientific computing in Python. Nature Methods 17:261–272. <https://doi.org/10.1038/s41592-019-0686-2>
629

- 630 [33] Wald DJ, Dewey JW (2005) Did You Feel It? U.S. Geological Survey data release,
631 <https://doi.org/10.5066/F7J101C8>
- 632 [34] Wald DJ, Quitoriano V, Dengler LA, et al (1999) Utilization of the Internet for
633 rapid community intensity maps. *Seismological Research Letters* 70(6):680–697.
634 <https://doi.org/10.1785/gssrl.70.6.680>
- 635 [35] Wald DJ, Quitoriano V, Worden CB, et al (2012) USGS “Did you feel it?”
636 internet-based macroseismic intensity maps. *Annals of Geophysics* 54(6):688–707.
637 <https://doi.org/10.4401/ag-5354>
- 638 [36] Wilkinson MD, Dumontier M, Aalbersberg IJ, et al (2016) The FAIR guid-
639 ing principles for scientific data management and stewardship. *Scientific Data*
640 3:160018. <https://doi.org/10.1038/sdata.2016.18>